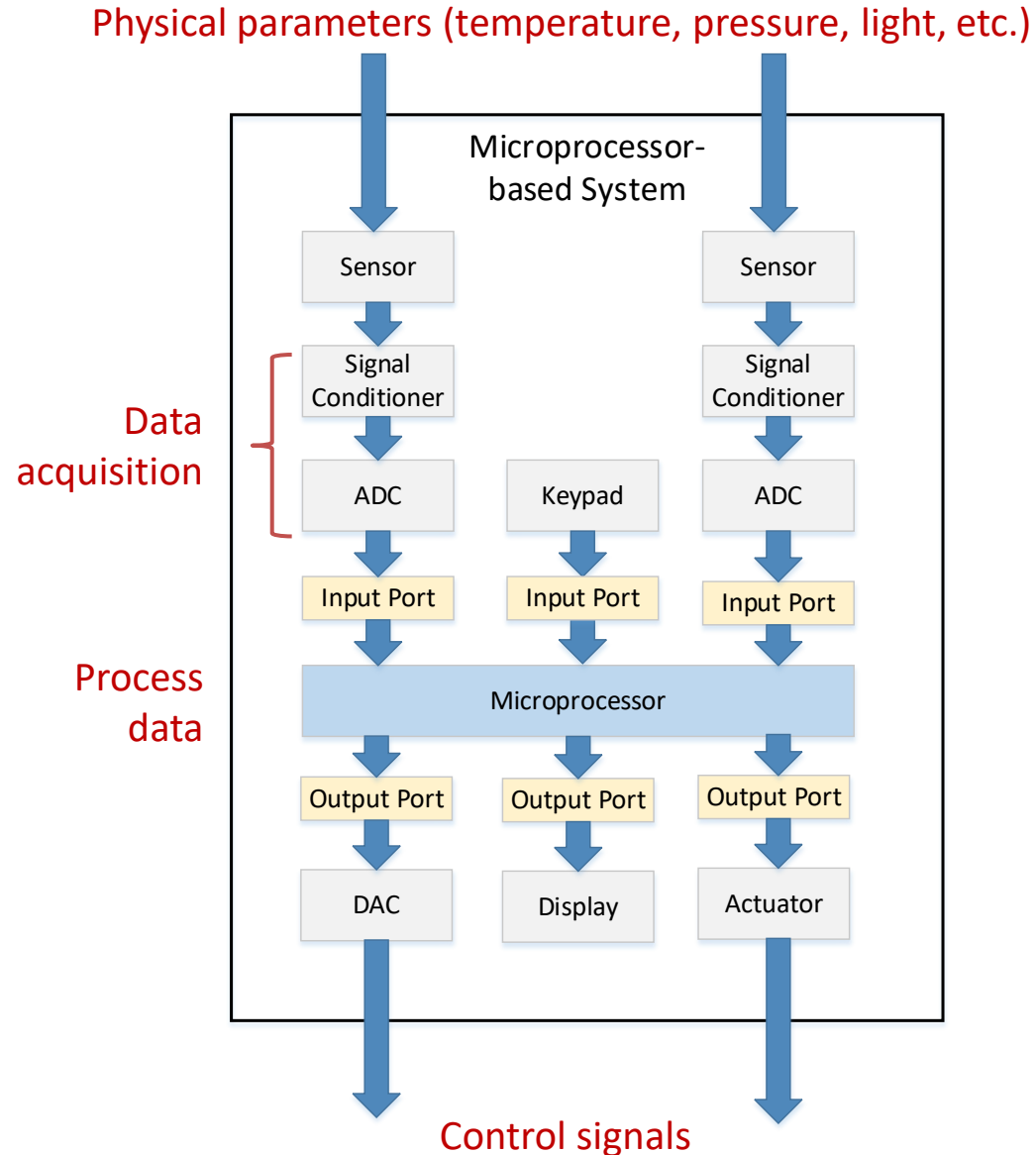


Recap

- Typical functions of an embedded system for automated control:
 - Monitor one or few physical parameters (temperature, pressure, light) and data acquisition
 - Process the acquired data
 - Control the physical parameters - system needs to generate signals



CE2007

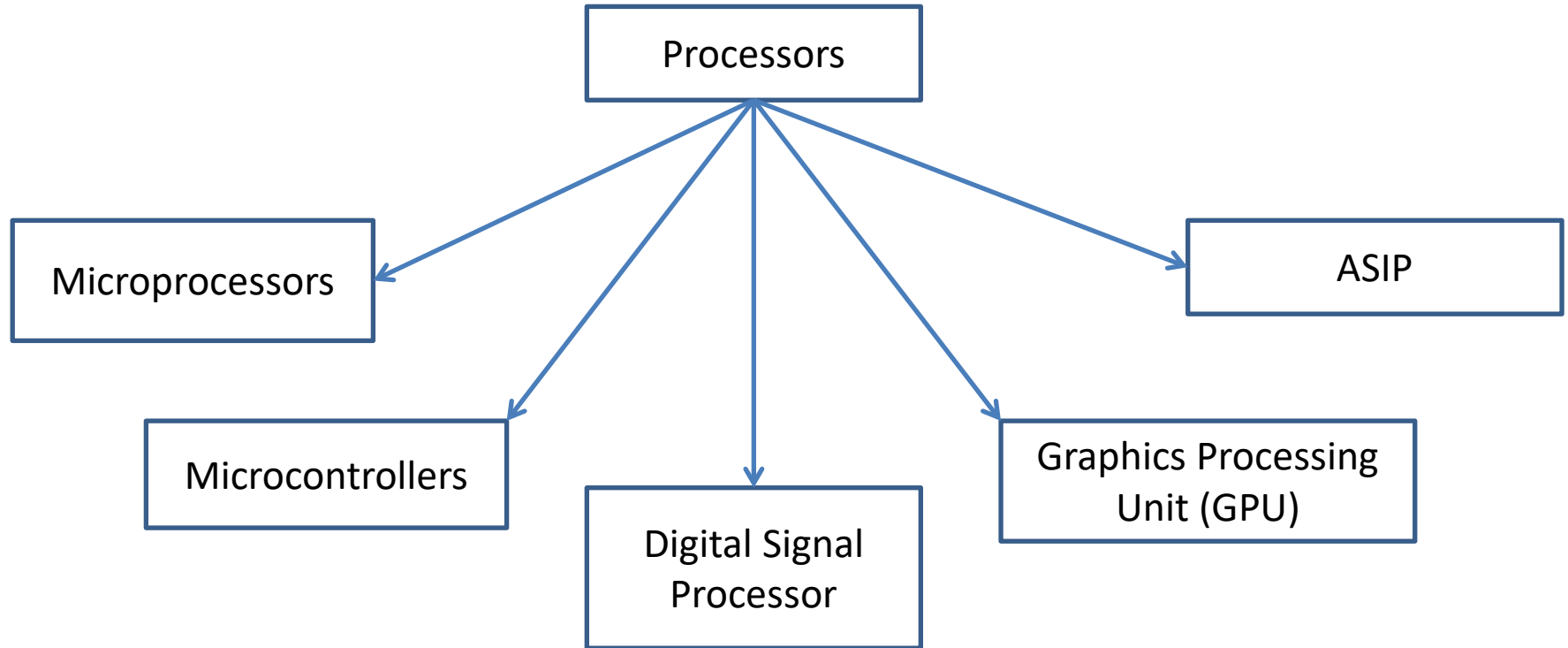
Microprocessor-based System Design

Microprocessor Landscape

Topics Covered

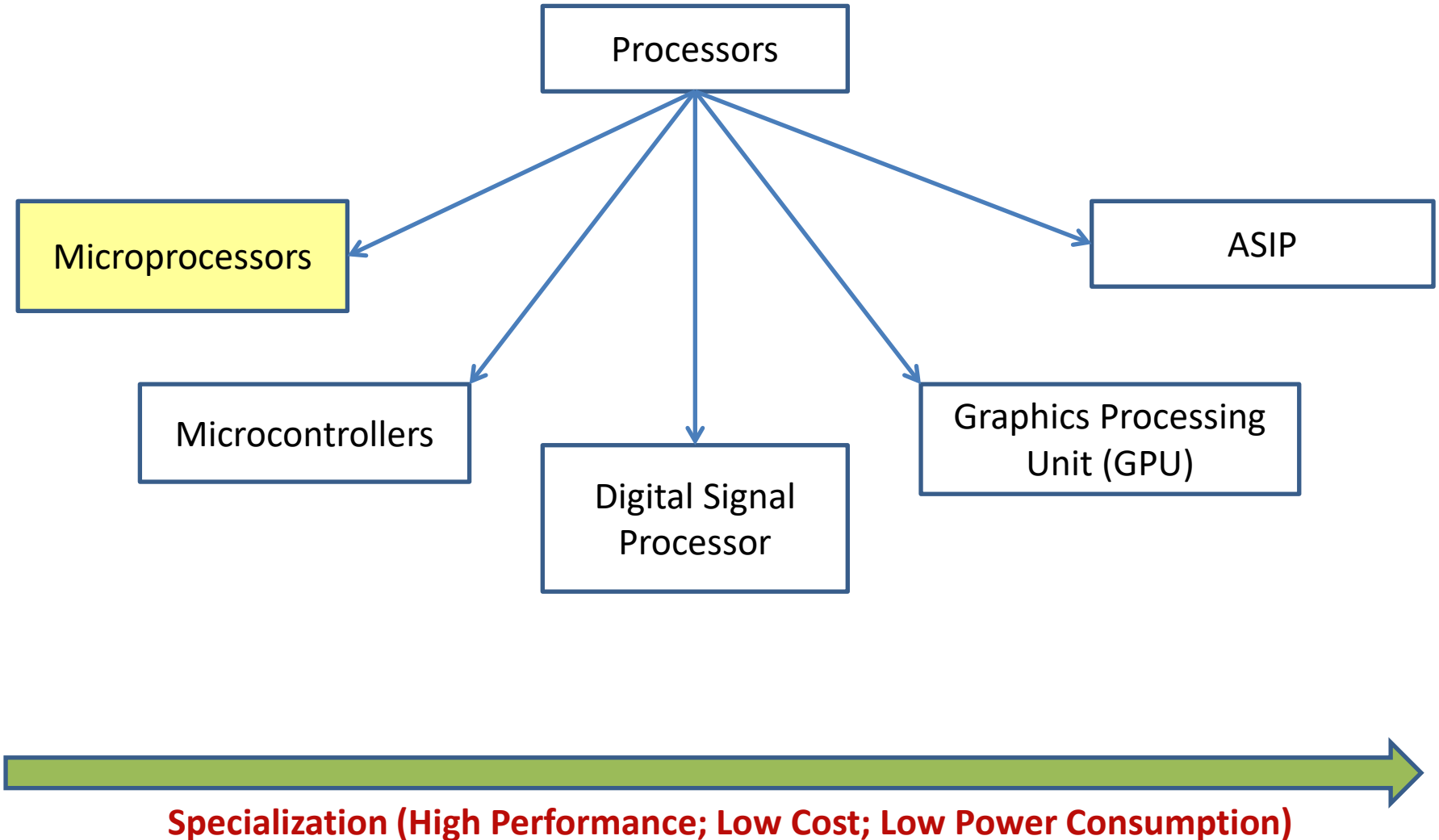
- Classification of Processors
- Microprocessors
- Microcontrollers
- Digital Signal Processors
- Introduction to ARM

Classification of Processors



Specialization (High Performance; Low Cost; Low Power Consumption)

Classification of Processors



Microprocessor - Definition

- Microprocessor refers to a **single chip implementation** of a computer's processor, or Central Processing Unit (CPU)
 - i.e. a CPU on a single Integrated Circuit (IC)
- CPU **fetches** and **executes** a set of **general-purpose instructions**
 - E.g. Data transfer operations, ALU operations, stack operations, I/O operations, program control, etc.
- In order to implement a fully operating computing system, additional resources such as **memory**, **I/O peripherals**, etc. must be **interfaced** with the microprocessor

Microprocessor - Applications

- Today, microprocessors are typically used for computing systems that are design to perform **multiple functions** (i.e. general purpose e.g. PC or server machines)
- Microprocessors are used for the following reasons:
 - High-speed and complex calculations are required while supporting multiple users (e.g. PCs, servers or super-computers)
 - Ready availability of compiler tools to facilitate development in high-level language
 - Ready availability of well-tested APIs and application codes (high reusability)

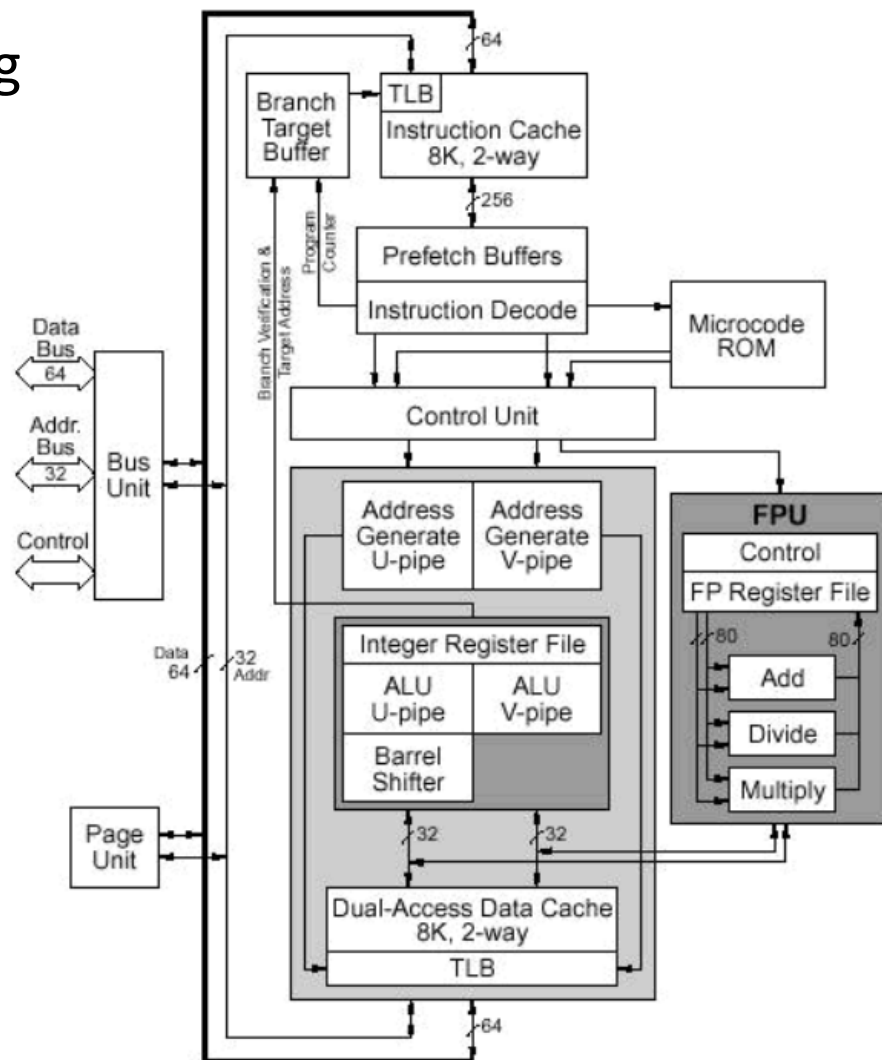


Personal Computers

Source: http://www.upandrun.co.uk/product_pages/pcs/pcs.html

Microprocessor – Typical Features

- Wide address bus for accessing large memory spaces
- Wide data formats (32-bits or more)
- Arithmetic Logic Unit (ALU) with integrated co-processors (e.g. Floating-Point Unit (FPU))
- Sophisticated addressing modes to efficiently support high-level languages
- Large silicon area (High cost)
- High power consumption



Architecture of Intel Pentium microprocessor

Source: <http://www.laynetworks.com/Block%20diagram%20of%20the%20Pentium.htm>

Characteristics of a Microprocessor

- There are numerous ways for designing microprocessors
 - Typically influenced by certain requirements (e.g. speed, power, size)
- Hence, there are many ways to characterize a microprocessor
 - RISC or CISC
 - Von-Neumann or Harvard
 - Bit size
- The most basic characteristic that is used to describe a microprocessor is probably the:
 - Bit size of a microprocessor (e.g. 8-bit, 32-bit, 64-bit)

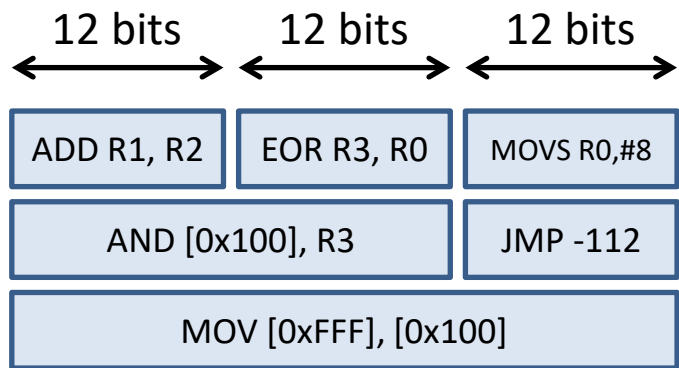
Question 1

Which of the following does the bit size of a microprocessor refers to (there can be more than one answer)?

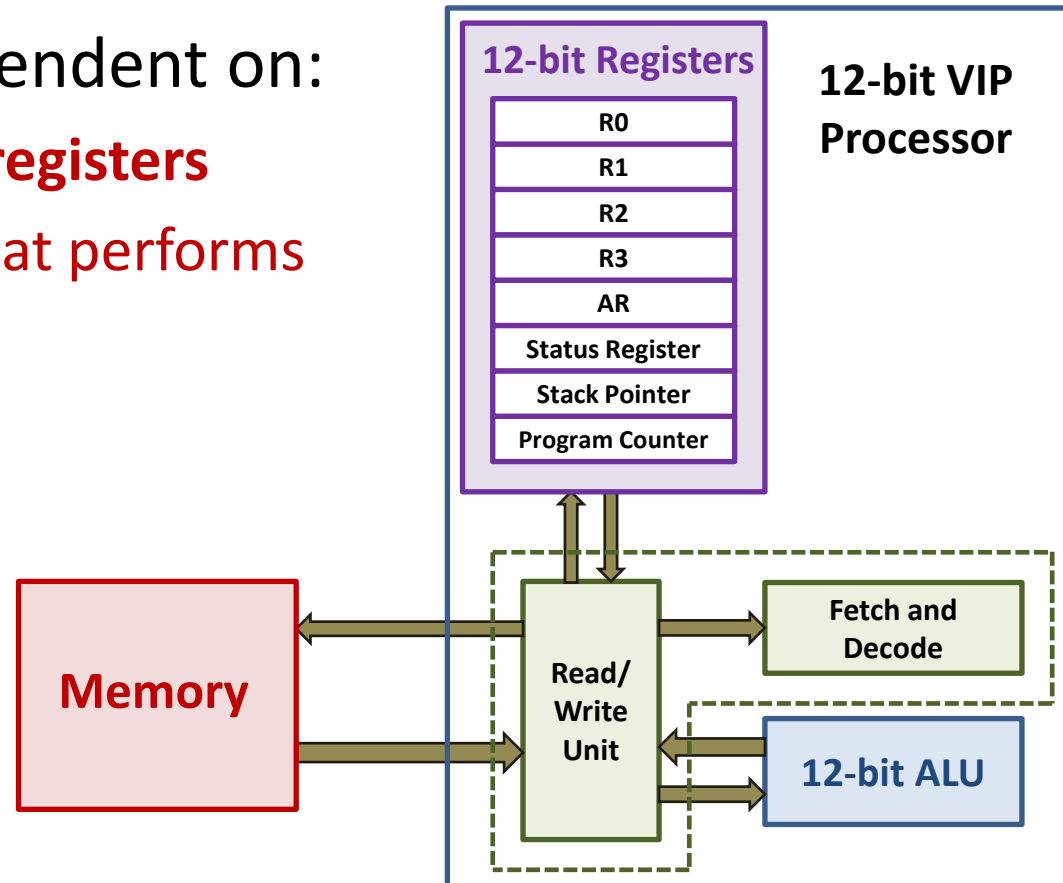
- A. Address bus width
- B. Data bus width
- C. Size of the data that can be operated at any one time
- D. Instruction Length
- E. Register Size

Microprocessor's Bit Size

- The **size of the data** that **each instruction can operate at any one time**
 - Note that the instruction could be of variable length
- Bit size is directly dependent on:
 - Size of microprocessor **registers**
 - In particular, the **ALU** that performs the operations



Variable length VIP Instructions

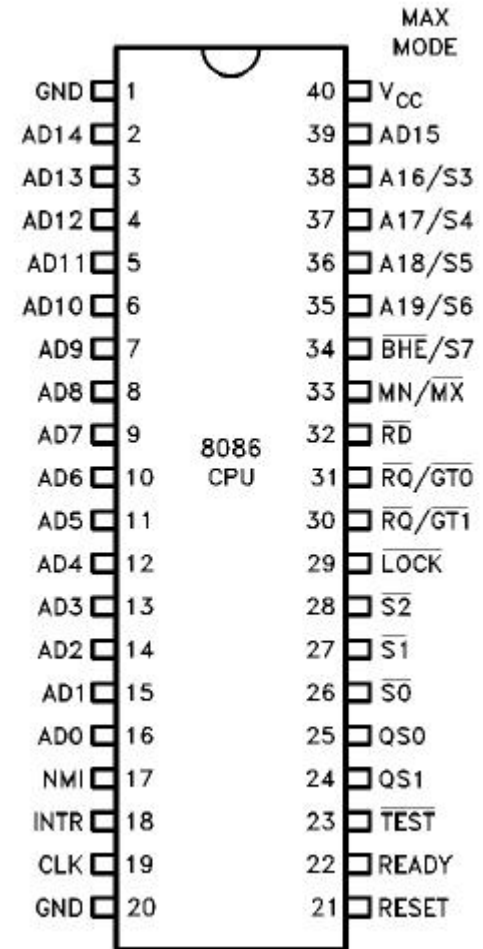
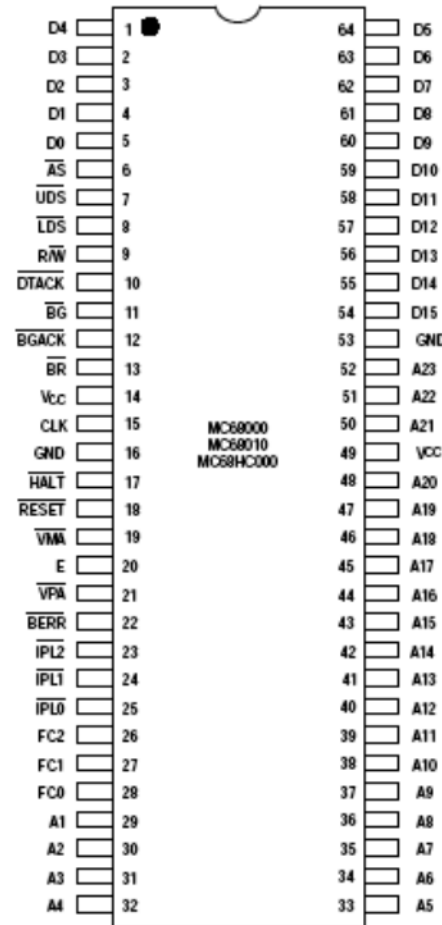


Address Bus Width

- The width (size) of a processor's address bus is typically **not** the defining factor for a processor's bit size, example:

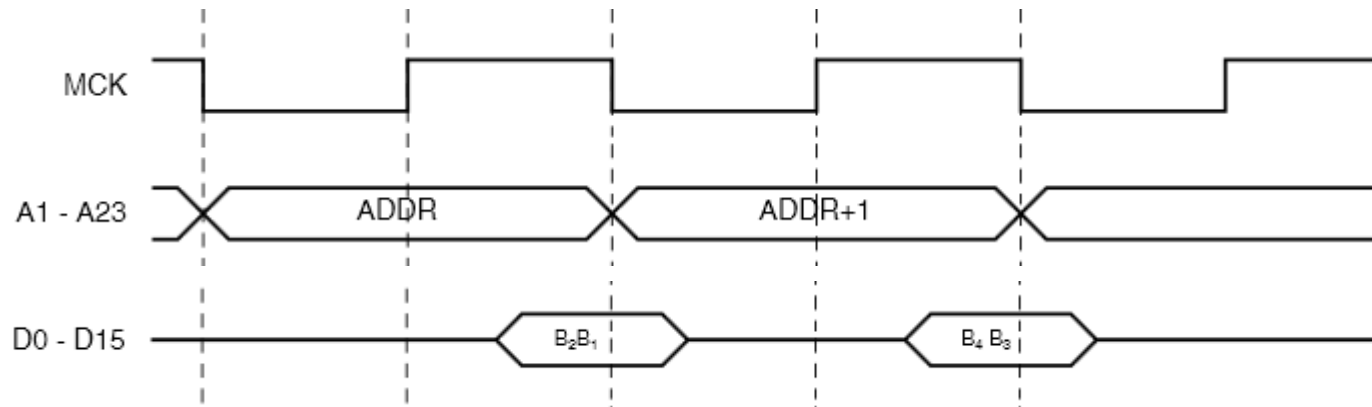
- 32-bit 68000 has an address bus of only 24 bits
- 16-bit 8086 uses segment register addressing to get 20 address bits

- Many 8-bit CPUs (e.g. 8051) implements 16-bit addressing (otherwise the CPU can only access 256 memory locations)



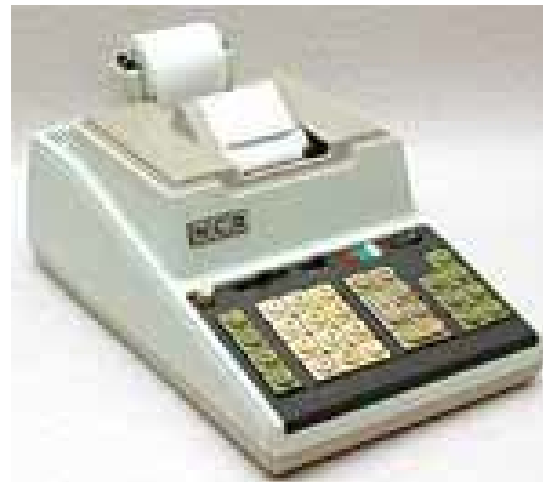
Data Bus Width

- The processor's **external** data bus width (size) also **may not** always properly reflect the processor's bit-size, example:
 - A 32-bit ARM processor could use 8-bit or 16-bit external data bus when accessing SDRAM or Flash memory
 - A 32-bit data will hence be transferred in two data bus cycles for a 16-bit data external bus system

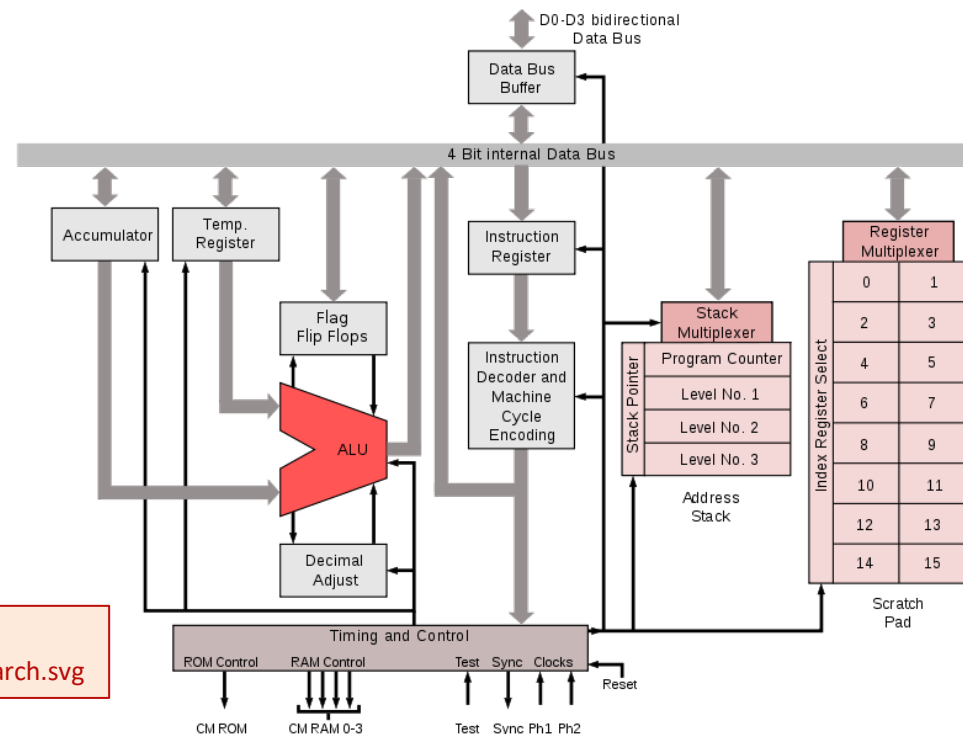


4-bit Microprocessors

- The first commercial microprocessor was the 4-bit Intel 4004
 - Developed for Busicom print calculator in 1971
 - Binary-Coded Decimal (BCD) based 4-bit ALU
 - 8-bit instructions and 12-bit addressing



The calculator that spawned the microprocessor
Source:
http://www.vintagecalculators.com/html/busicom_141-pf_and_intel_4004.html



Intel 4004 Architecture
Source: http://en.wikipedia.org/wiki/File:4004_arch.svg

4-bit Microprocessors – Applications and Examples

- Nowadays, 4-bit microprocessors are typically implemented in the form of microcontrollers and used in low cost, high volume applications such as simple control and monitoring systems
 - Watches and clocks, toys, TV/HiFi remote controls, air conditioning, microwave ovens, refrigerators, and many other consumer electronic devices
- Examples of 4-bit Microprocessors
 - Atmel MARC4
 - Targeted for low voltage/power application e.g. hand-held and battery operated devices/equipment
 - EM Microelectronic EM6607
 - Low power, mask programmed CMOS 4-bit microcontroller
 - Epson S1C60 Series
 - For small sized equipment e.g. clock, timers, thermometers

8-bit Microprocessors

- 8-bit microprocessors implemented in single chip **microcontrollers**, are still the most widely used processors
- Advantages:
 - 8-bit processors matches byte-size data naturally
 - Memories are usually byte-addressable
- Typically uses 16-bit address bus to provide sufficient memory space for practical applications
- Most widely used 8-bit processor is probably the **Intel 8051** (1980 Harvard architecture microcontroller)
 - Many variants of the 8051 are still produced today
- Examples of modern 8-bit microcontrollers
 - Microchip's PIC microcontroller
 - Atmel's AVR family microcontroller

16-bit Microprocessors

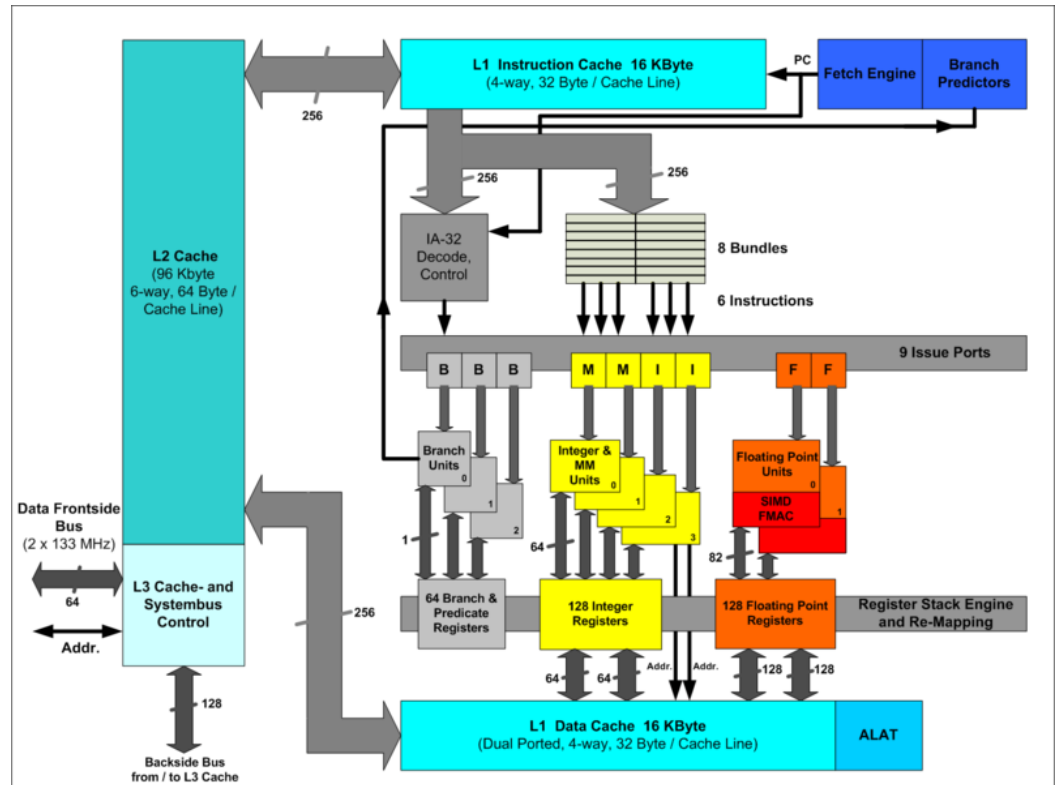
- Examples:
 - Intel 8086 (origin of the x86 Instruction Set Architecture for PC today)
 - Texas Instruments MSP430
 - Microchip XLP series
 - Seems to be popular in processors designed by Japanese companies
- Applications
 - Smart cards, PC peripherals, digital camera
- Nowadays, 16-bit designs are commonly found in Digital Signal Processors (DSP) – more on this later

32-bit Microprocessors

- The first microprocessor with a 32-bit programming model is the Motorola 68000
- Well-known 32-bit processors are those produced by Intel for PCs
 - 80386, 80486, Pentium, etc.
- In the past, 32-bit processors are predominantly used in general purpose computer systems (e.g. PCs)
 - They are now superseded by the 64-bit processors (e.g. Intel iCore series)
- Today, most common 32-bit processors belong to the ARM processor families
 - Widely used for smart phones and tablets

64-bit Microprocessors

- With advances in semiconductor technology, it is now possible to implement 64-bit processors on a single silicon chip
- Intel currently has the highest market share for 64-bit processor
 - In 2012, Intel produced 80% of processors used in PCs
- 64-bit processors are most suitable for high-end computing systems (e.g. multi-user server)



Intel Itanium Architecture

Source: http://en.wikipedia.org/wiki/File:Itanium_arch.png

Question 2

- Which of the following do you think is the microprocessor in the first laptop?
 - A. 8-bit Zilog Z80
 - B. 16-bit Intel 8086
 - C. 32-bit ARM Cortex-M
 - D. 64-bit AMD Athlon

The First Laptop

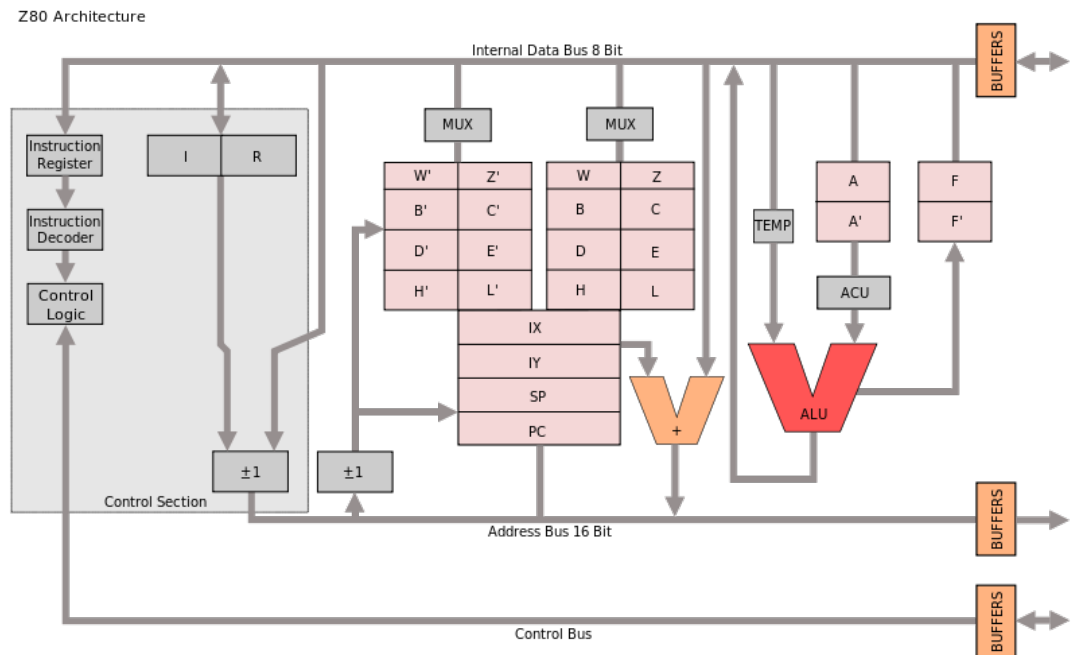
- In 1981, **Osborne Computer Corporation** released the first commercially successful portable microcomputer, called **Osborne 1**

- Weighed 10.7kg
- Cost USD \$1,795
- 5-inch display screen
- 64 KB Main Memory
- CPU: 4 MHz Zilog Z80



Osborne

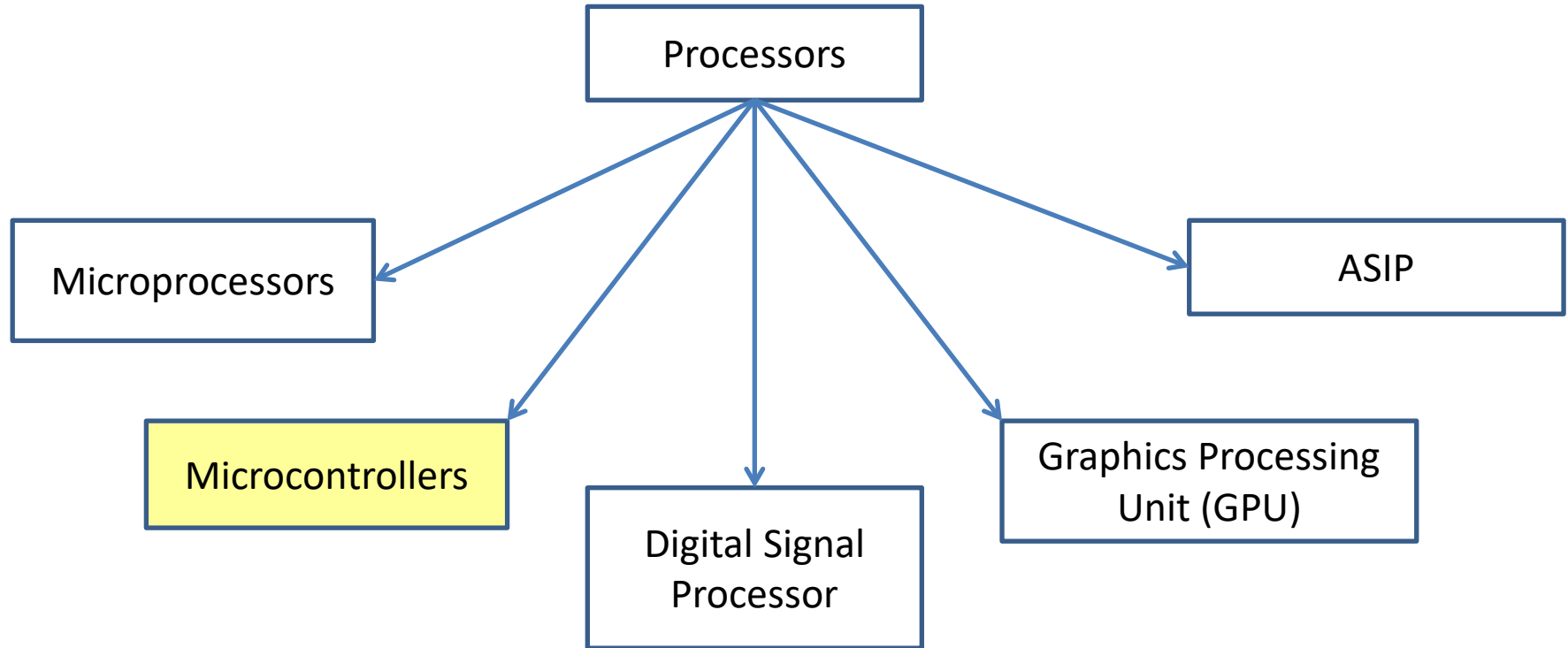
Source: http://en.wikipedia.org/wiki/Osborne_1



Z80 Architecture

Source: http://en.wikipedia.org/wiki/File:Z80_arch.svg

Classification of Processors



Specialization (High Performance; Low Cost; Low Power Consumption)

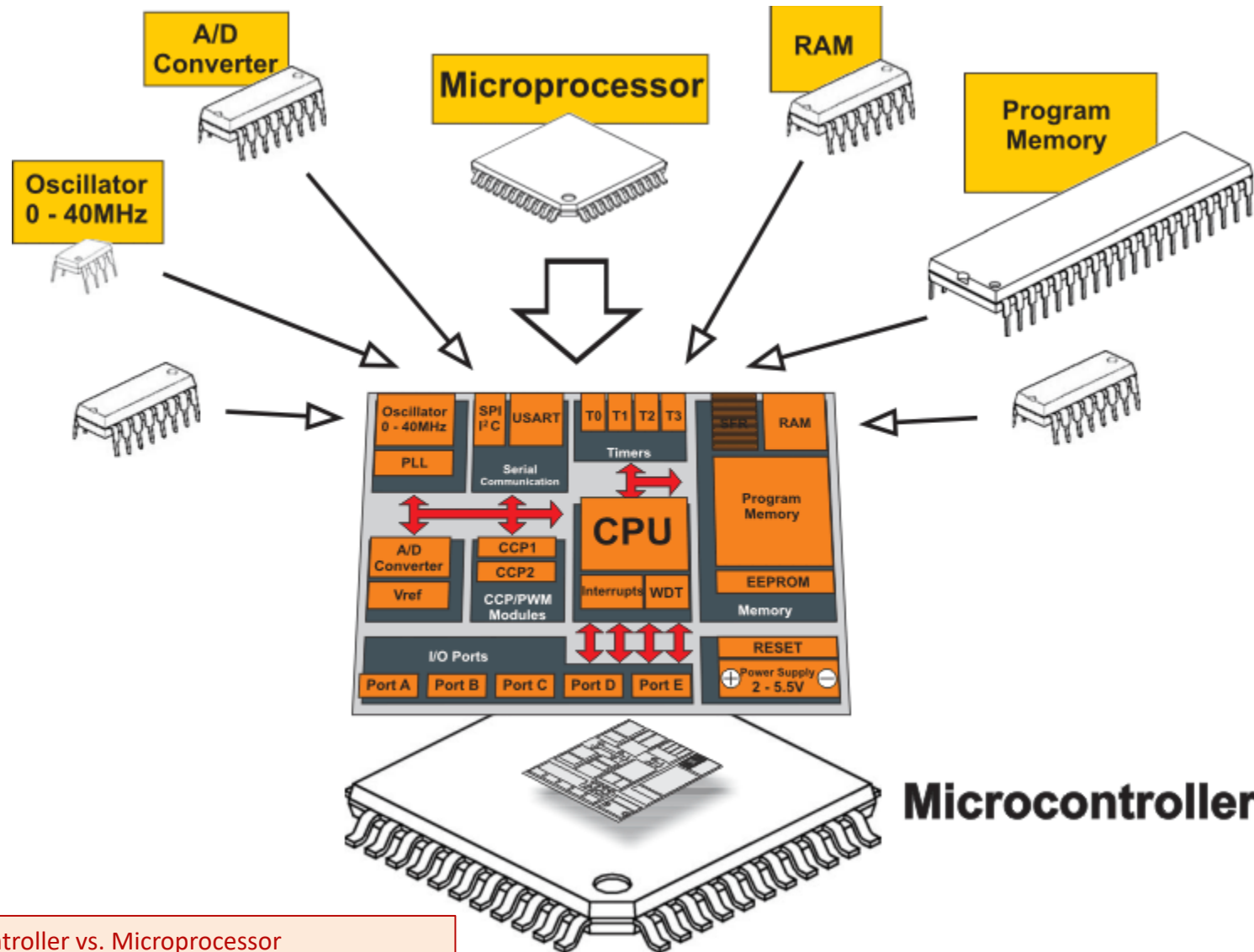
What are Microcontrollers?

- In order to communicate with external environment, the **microprocessor** must use specialized circuits added as external chips
- The **microcontroller** is designed to be **all of that in one**
 - No specialized external components are required for its application as all necessary circuits which otherwise belong to peripherals are already built into it
- Advantage of microcontrollers:
 - Saves time and space needed to design a device



NXP LPC1102 – 32-bit ARM microcontroller
Source: <http://www.radiant.su/eng/news/?action=show&id=144>

Microcontrollers vs. Microprocessor



Microcontroller vs. Microprocessor

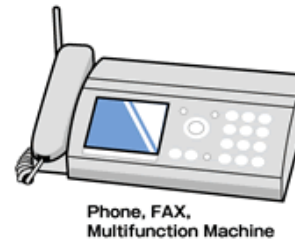
Source: <http://www.mikroe.com/chapters/view/1/>

Microcontroller - Applications

- Microcontrollers are used in **embedded systems** to perform **specific functions** (typically **control operations**)
 - Real-time response are required for certain applications
- Microcontrollers are used when part of the embedded software needs to reside in internal memory and when **on-chip functional units** (e.g. interrupt-handler, port, timer, ADC, etc.) are required

- Typical applications:

- Television set
- Wristwatches
- Remote control
- USB controller
- Computer printer
- Electronic fuel injection
- ABS brakes



Vehicle safety applications
Source: <http://www.infineon.com/cms/en/corporate/press/news/releases/2007/INFAIM200707-074.html>

General-purpose ARM Microcontroller family
Source: <http://www.lapis-semi.com/en/semicon/micon32/micon32.htm>

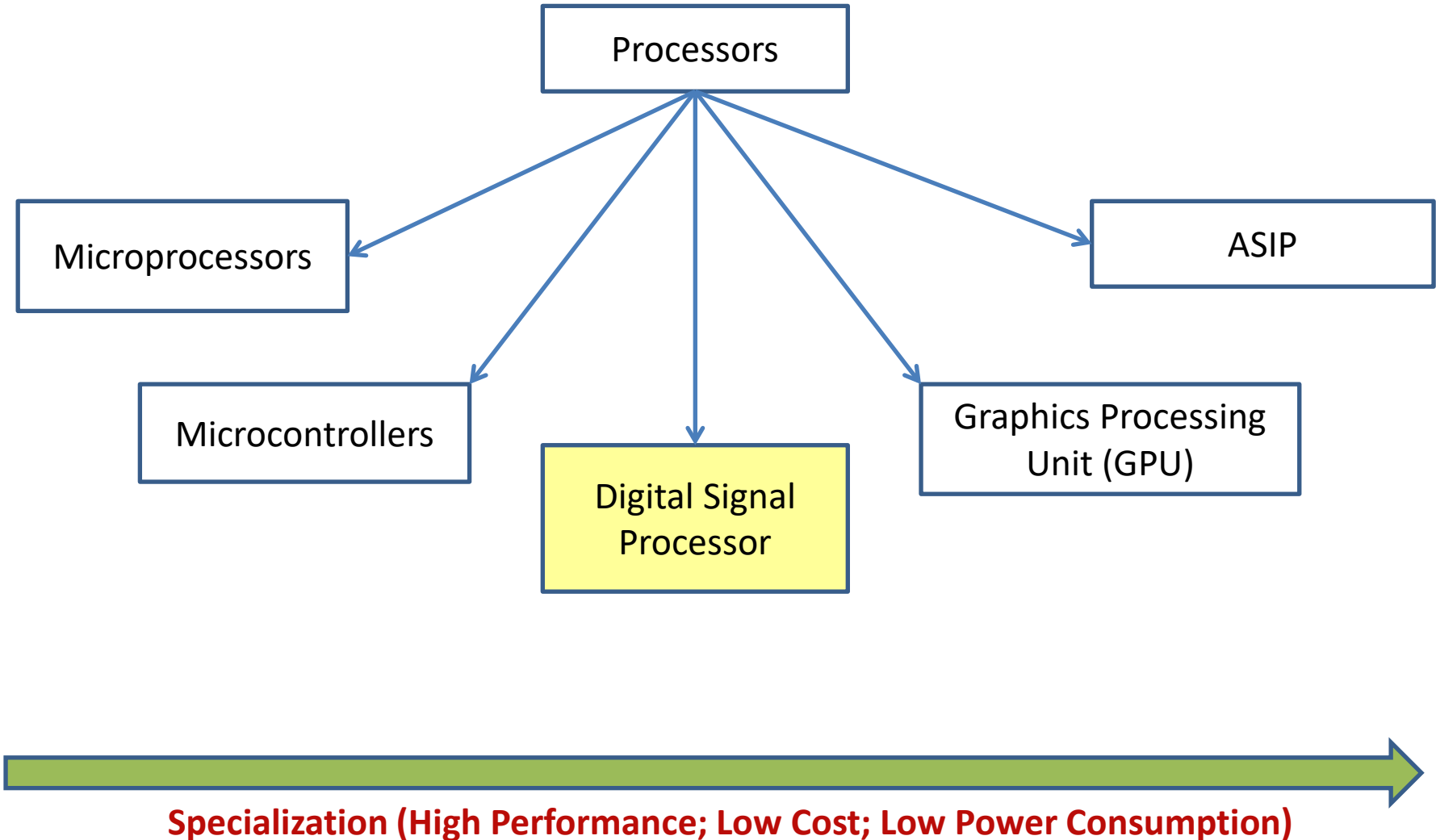
Microcontroller – Typical Features and Examples

- Memory and peripherals **integrated on chip**
- **Narrow address bus** allowing limited amount of memory
- **Narrow data formats** (typically 8-bits or 16-bits)
- **Limited** addressing modes (poor support for high-level language programming)
- **Low cost** and **Low power consumption**

- ARM core processors (many vendors)
 - ARM Cortex-M cores are specifically targeted towards microcontroller applications
- Atmel AVR (8-bit), AVR32 (32-bit), and AT91SAM (32-bit)
- Cypress Semiconductor's M8C Core used in their PSoC (Programmable System-on-Chip)
- Freescale ColdFire (32-bit) and S08 (8-bit)
- Freescale 68HC11 (8-bit)
- Intel 8051
- Infineon: 8bit XC800, 16bit XE166, 32bit XMC4000 (ARM based Cortex M4F), 32bit TriCore and , 32bit Aurix Tricore Bit microcontrollers^[9]
- MIPS
- Microchip Technology PIC, (8-bit PIC16, PIC18, 16-bit dsPIC33 / PIC24), (32-bit PIC32)
- NXP Semiconductors LPC1000, LPC2000, LPC3000, LPC4000 (32-bit), LPC900, LPC700 (8-bit)
- Parallax Propeller
- PowerPC ISE
- Rabbit 2000 (8-bit)
- Renesas Electronics: RL78 16-bit MCU; RX 32-bit MCU; SuperH; V850 32-bit MCU; H8; R8C 16-bit MCU
- Silicon Laboratories Pipelined 8-bit 8051 Microcontrollers and mixed-signal ARM-based 32-bit microcontrollers
- STMicroelectronics STM8 (8-bit), ST10 (16-bit) and STM32 (32-bit)
- Texas Instruments TI MSP430 (16-bit) C2000 (32-bit)
- Toshiba TLCS-870 (8-bit/16-bit).

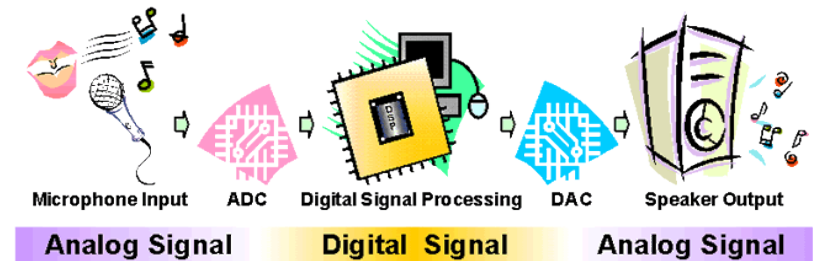
Microcontroller architectures and vendors
Source: <http://en.wikipedia.org/wiki/Microcontroller>

Classification of Processors



Digital Signal Processing

- Many signal processing functions (such as filtering and convolution operations) are now performed in the digital domain using **digital signal processing algorithms**
- These algorithms typically require a large number of mathematical operations to be performed **quickly** (real-time) and **repeatedly** on a series of data samples.
- Full features of a **general purpose processor** (e.g. operating system support) are **not** required to execute these digital signal processing algorithms



Audio signal processing

Source:

<http://macao.communications.museum/eng/exhibition/secondfloor/moreinfo/adconverter.html>

Digital Signal Processors - Applications

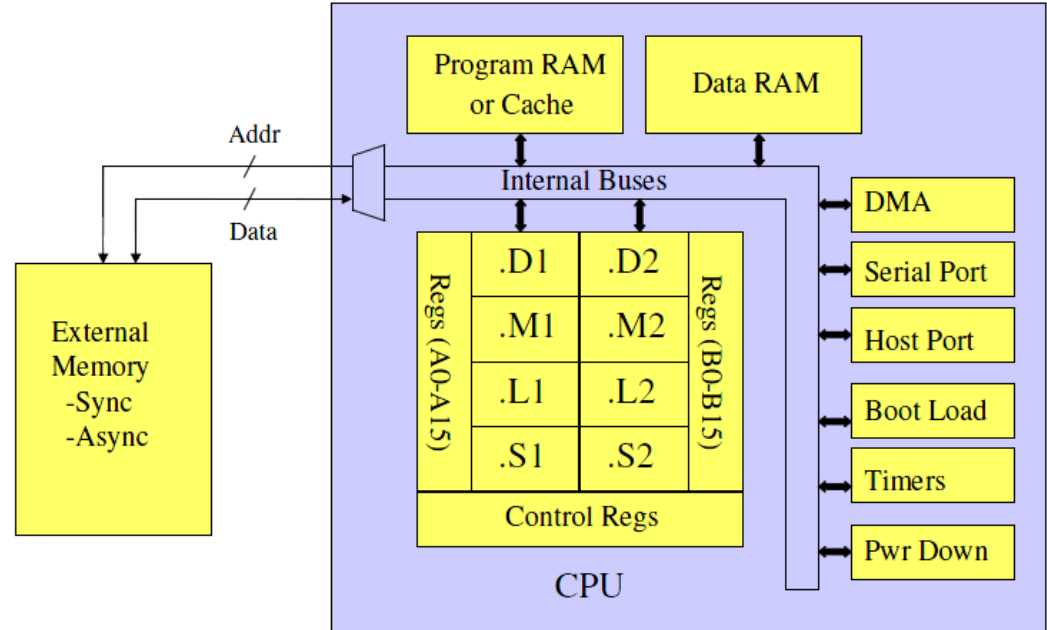
- Digital Signal Processor (DSP) is used when signal processing functions are to be processed fast
- Typical applications:
 - Telecommunication systems
 - Flight control systems in aerospace applications
 - Audio/video applications in MP3 players, video cameras
 - Modems
 - Noise cancellation

Digital Signal Processor – Typical Features

- Narrow address bus supporting limited amount of memory
- Narrow data formats (typically 16-bits or 32-bits)
- Low cost and Low power consumption
- Specialized addressing modes to support signal processing operations
 - Modulo addressing, bit-reverse addressing
- Architecture optimized for intensive computations (e.g. Multiply-Accumulate (MAC units))
 - Many fundamental digital signal processing algorithms (e.g. FIR filters or FFT) depend heavily on MAC operations

Digital Signal Processor – Examples

- Examples of leading DSP manufacturers
 - Texas Instruments Inc.
 - C6000 series DSP
 - Analog Devices Inc.
 - SHARC-based DSP
 - Freescale Semiconductor Inc.
 - MSC81xx multi-core DSP family



Two parallel data paths

- ▶ Data unit - 32-bit address calculations (modulo, linear)
- ▶ Multiplier unit - 16 bit $\times$ 16 bit with 32-bit result
- ▶ Logical unit - 40-bit (saturation) arithmetic & compares
- ▶ Shifter unit - 32-bit integer ALU and 40-bit shifter

Introduction to the TMS320C6000 VLIW DSP

Source: <http://www.docstoc.com/docs/2684927/INTRODUCTION-TO-THE-TMS320C6000-VLIW-DSP>

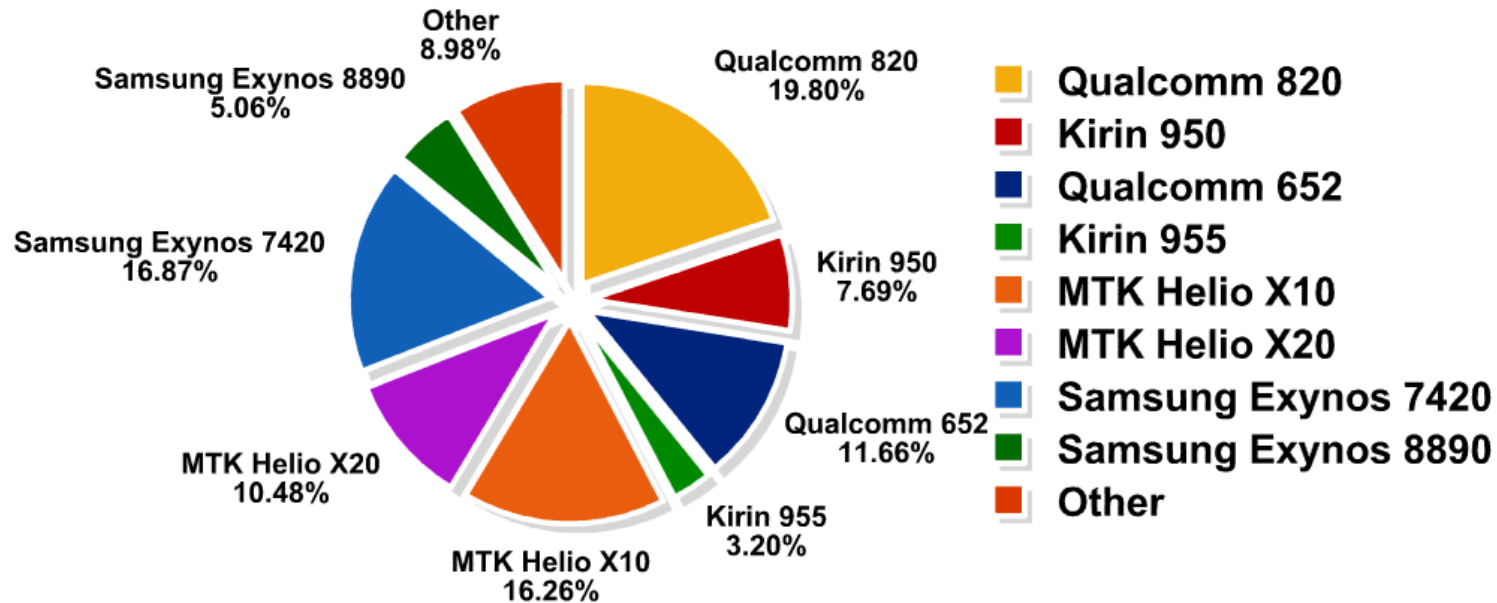
Question 3

- Why microcontrollers are not suitable to run DSP algorithms?
 - A. Microcontrollers do not support large memories
 - B. Microcontrollers are expensive
 - C. Microcontrollers do not have specialized architectures for DSP operations
 - D. Microcontrollers consumes low power

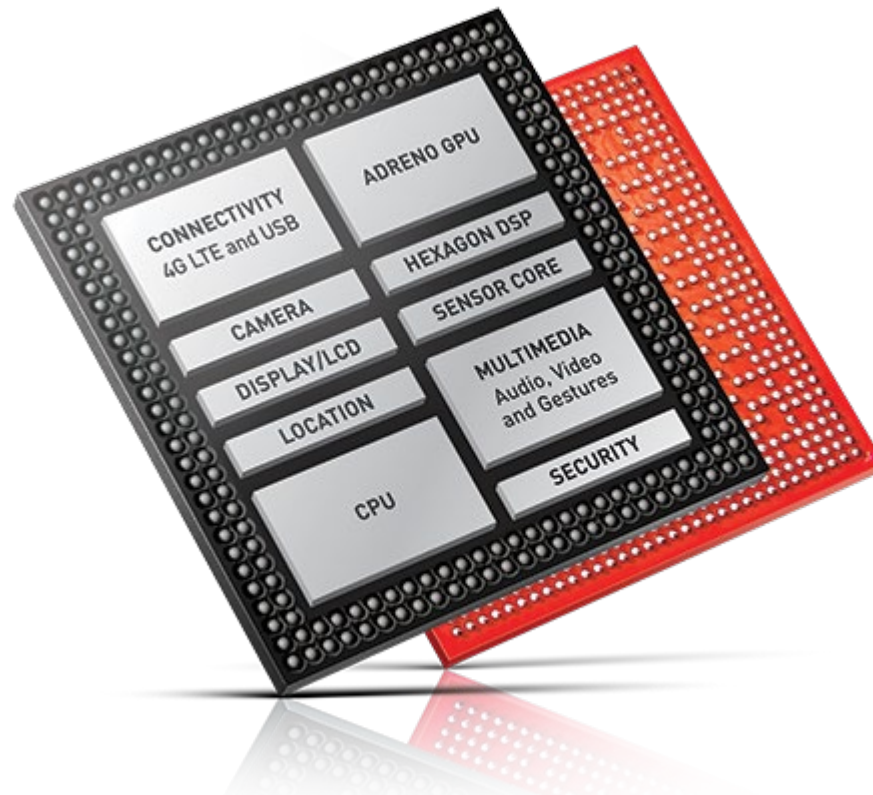
Market Share of Smart Phone CPU

Distribution of Popular Smart Phone CPU, 1H 2016

Data Sources: AnTuTu Benchmark (2016.01-06)



Qualcomm Snapdragon



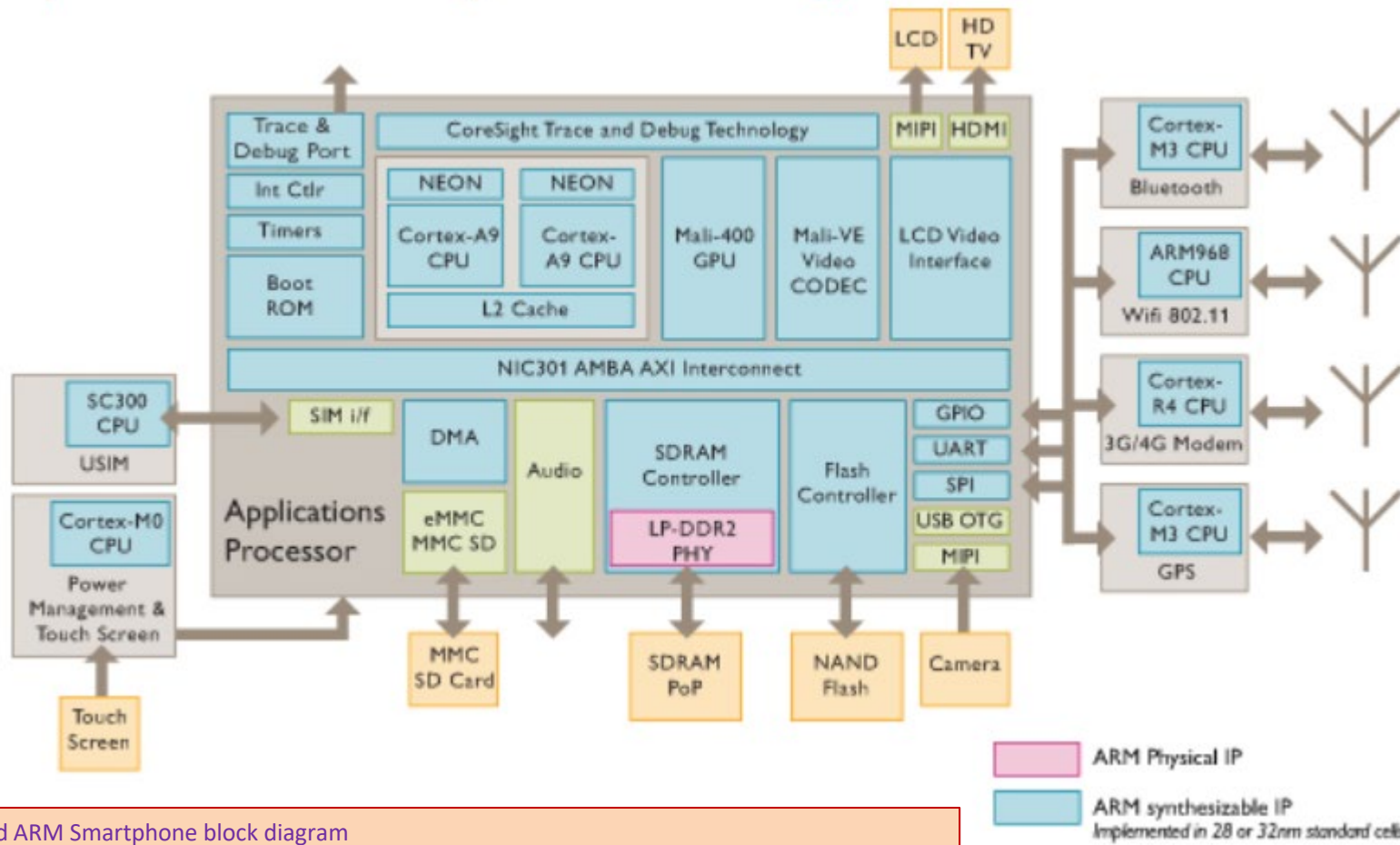
Source: <https://www.qualcomm.com/products/snapdragon/processors/810>

Qualcomm® Snapdragon™ 810 processor with 64-bit computing on 8 CPU cores

ARM Processor

- Currently, the most popular processor is ARM

Optimized ARM Smartphone Block Diagram



Optimized ARM Smartphone block diagram

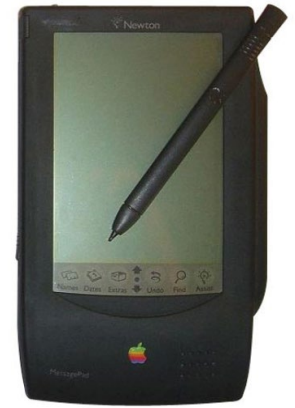
Source: <http://www.arm.com/markets/mobile/smartphones.php>

A Short History of ARM

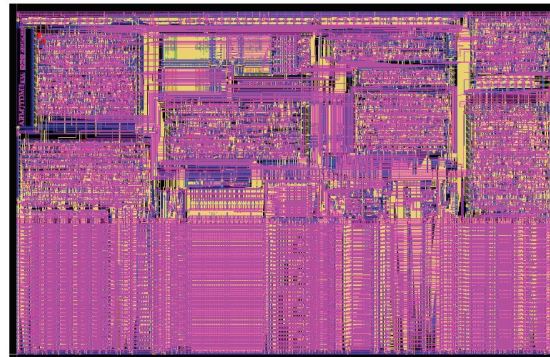
- **ARM** was founded in 1990 in Cambridge, UK as a joint venture between Acorn Computers, Apple Computer and VLSI Technology
- Goal was to design **32-bit processors** to meet **higher functionality** and **lower power** requirements in **embedded systems**
- ARM stands for **Advanced RISC Machine**
 - Based on Reduced Instruction Set Computers (RISC) architectures
 - Trades simpler (and hence faster) hardware circuitry with software complexity (and size)
 - Today's ARM architectures are enhanced versions of RISC
 - Latest ARM processors has more than 100 instructions

A Short History of ARM (Cont.)

- First significant visibility when Apple introduced its Newton PDA (1993) that used an ARM6 core – connectivity was a problem
- About the same time, cellphone makers were interested in more powerful processors but the 32-bit RISC processor produces large code size
 - ARM found a way to encode its 32-bit instructions to a 16-bit instruction set (**THUMB**)
 - The **ARM7TDMI** processor was born in 1992 and became very popular in mobile phones



Apple Newton PDA



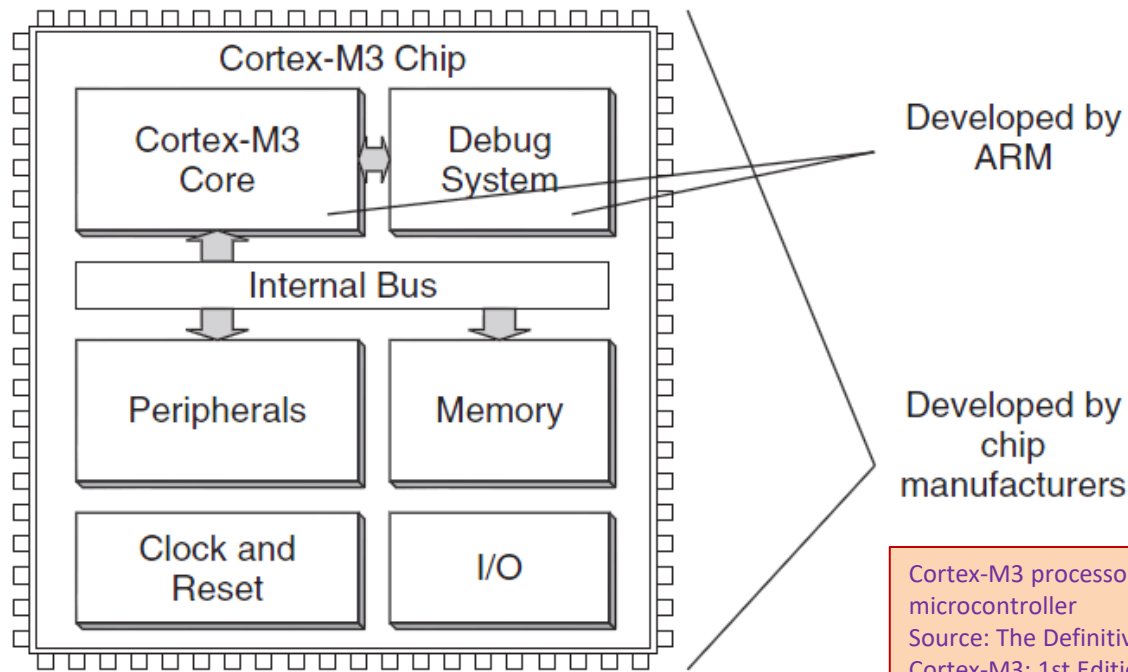
Small, low power
32-bit core for
wireless baseband

Early GSM phone



ARM Business Model

- ARM **never** manufacture their processors
 - Sold under soft or hard IP core license
 - Licensees known as ARM partners
 - Qualcomm, Apple, Samsung, Intel, Microsoft
- Licensees add their own logic and customized peripherals, and manufacture the processors



Cortex-M3 processor vs. Cortex-M3 based microcontroller
Source: The Definitive Guide to the ARM Cortex-M3; 1st Edition; Joseph Yiu; 2007

ARM's big.LITTLE Processor

- Heterogeneous computing architecture that combines a Cortex-A15 (big) and Cortex-A7 (LITTLE) processors on a single die
- A15 run when computing demands are high, otherwise it sleeps and A7 runs exactly the same code while consuming less power
- Example: A smart phone needs high computational power when fully active, but spends most of it's time in idling mode



Big.LITTLE processing

Source: <http://www.arm.com/products/processors/technologies/biglittlprocessing.php>

ARM Processor Families

■ Classic

- ARM7, ARM9, ARM11

■ Cortex

- Cortex-A, Cortex-R, Cortex-M

Multicore

ARMv7-A

Cortex-A15

ARMv7-A

Cortex-A9

Cortex-A8

ARMv7-R

Cortex-R4

ARM1176,
ARM1156

ARMv7-M

Cortex-M3

ARM1136

ARM926EJ-S

ARMv4T

ARM7TDMI-S,
ARM920,
ARM10 ARM9E

ARM922T

ARM9TDMI

ARM7500

ARM7100

ARM7

ARMv6-M

Cortex-M0

Cortex-M4

Cortex-M1

